

GROUP PROJECT
OF
DATA ACQUISITION & SYSTEM DESIGN
(UEI701)

COURSE INSTRUCTOR:
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1. Aim:

To design and implement a Soil Moisture Data Acquisition System using Arduino Uno, a soil moisture sensor, 16×2 LCD display with I2C module, and LED indicators to measure, display, and indicate soil moisture level in real time.

2. Apparatus Required:

1. Arduino Uno – 1
2. Soil Moisture Sensor Module – 1
3. 16×2 LCD Display – 1
4. I2C Module for LCD – 1
5. Breadboard – 1
6. Jumper Wires (Male-to-Male, Male-to-Female) – As required
7. USB Cable for Arduino – 1
8. Red LED – 1
9. Green LED – 1
10. 220Ω Resistors – 2
11. Computer/Laptop with Arduino IDE installed

3. Theory:

3.1 Arduino Uno

The Arduino Uno is a microcontroller board based on the ATmega328P. It contains 14 digital I/O pins and 6 analog input pins. In this project:

- The analog pin (A0) reads moisture sensor data.
- Digital pins control LEDs.
- I2C communication (A4 – SDA, A5 – SCL) drives the LCD.

It converts analog signals (0–5V) into digital values (0–1023) using its 10-bit ADC (Analog to Digital Converter).

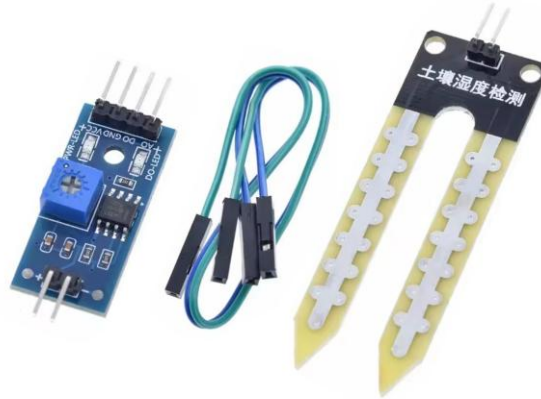


3.2 Soil Moisture Sensor

The soil moisture sensor measures the water content in soil. It works on the principle of electrical conductivity:

- Wet soil → High conductivity → Higher analog output.
- Dry soil → Low conductivity → Lower analog output.

The sensor provides an analog voltage proportional to soil moisture, which is read by the Arduino.



3.3 16×2 LCD Display with I2C Module

A 16×2 LCD can display 16 characters per line and has 2 lines.

The I2C module reduces the number of required connections by using only:

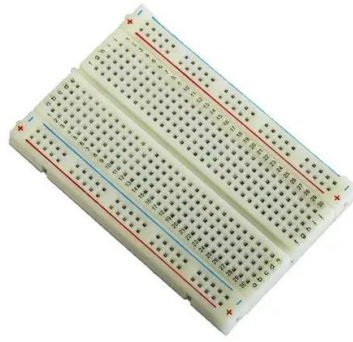
- SDA (Serial Data Line)
- SCL (Serial Clock Line)

Instead of using 6–8 data pins, only 2 pins are used, making the circuit simpler and more efficient.



3.4 Breadboard

A breadboard is used for temporary circuit connections without soldering. It helps in testing and prototyping the system.



3.5 Jumper Wires

Jumper wires are used to connect components on the breadboard to the Arduino pins.



3.6 USB Cable

The USB cable provides:

- Power supply (5V)
- Program uploading from computer to Arduino

3.7 LEDs (Red and Green)

- Red LED → Indicates dry soil.
- Green LED → Indicates sufficient moisture.

3.8 220 Ω Resistor

The 220 Ω resistor limits the current flowing through the LED to prevent damage.



Using Ohm's Law:

$$R = \frac{V}{I}$$

The resistor ensures safe operation of LEDs at 5V supply.

4. Procedure:

1. Connect the soil moisture sensor:
 - VCC → 5V
 - GND → GND
 - AO → A0
2. Connect LCD with I2C:
 - VCC → 5V
 - GND → GND
 - SDA → A4
 - SCL → A5
3. Connect LEDs:
 - Red LED → Pin 8 (through 220Ω resistor)
 - Green LED → Pin 9 (through 220Ω resistor)
 - Cathode → GND
4. Connect Arduino to computer using USB cable.
5. Open Arduino IDE.
6. Write and upload the program.
7. Insert sensor probe into soil.
8. Observe moisture value and LED indication.

5. Algorithm:

1. Start.
2. Initialize LCD and set pin modes.
3. Read analog value from moisture sensor.
4. Display moisture value on LCD.
5. Compare value with threshold.

6. If value < threshold:
 - Turn ON Red LED.
 - Turn OFF Green LED.
7. Else:
 - Turn ON Green LED.
 - Turn OFF Red LED.
8. Wait for 1 second.
9. Repeat from Step 3.
10. Stop.

6. Arduino Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
int sensorPin = A0;
int ledPin = 9; // PWM pin
```

```
void setup()
{
  pinMode(ledPin, OUTPUT);

  lcd.init();
  lcd.backlight();
  Serial.begin(9600);
}
```

```
void loop()
{
  int moistureValue = analogRead(sensorPin);

  // Convert to percentage
  int moisturePercent = map(moistureValue, 1023, 0, 0, 100);

  // Convert percentage to LED brightness (0–255)
  int brightness = map(moisturePercent, 0, 100, 0, 255);

  analogWrite(ledPin, brightness);
```

```
lcd.clear();
lcd.setCursor(0,0);
lcd.print("Moisture: ");
lcd.print(moisturePercent);
lcd.print("%");

lcd.setCursor(0,1);

if(moisturePercent <= 30)
{
  lcd.print("Status: DRY");
}
else if(moisturePercent <= 70)
{
  lcd.print("Status: MODERATE");
}
else
{
  lcd.print("Status: WET");
}

Serial.print("Moisture: ");
Serial.println(moisturePercent);

delay(800);
}
```

7. Conclusion:

The Soil Moisture Data Acquisition System was successfully designed and implemented using Arduino Uno. The system accurately measures soil moisture levels and displays the readings on a 16×2 LCD using I2C communication. LED indicators provide a simple visual representation of soil condition.

This system is low-cost, reliable, and suitable for agricultural and irrigation monitoring applications. It can be further enhanced by integrating automatic irrigation control or IoT-based remote monitoring.

